

ASSIGNMENT

Q1. What is SQL?

SQL stands for “*Structured Query Language*”. It is a standard Programming language used to communicate with the “*Relational databases*” in the form of tables. It helps users create, store, retrieve, update and delete data. SQL queries are mainly divided into four categories DDL, DQL, DML and DCL. SQL is also used to create tables, define relationships, and control access to databases. It is supported by popular and base systems such as *MySQL, Oracle, PostgreSQL, and SQL Server*.

Q2. What is the difference between Data & Information?

- **Data** consists of raw facts, figures, or observations that have not been processed. On its own, data may not have any clear meaning.
- **Information** is data that has been processed, organized, and Presented in a meaningful way. Information helps people understand situations and make better decisions. For example, exam marks are data, while the class average or result report is information.

Data	Information
Data is the raw, unorganized facts & figures.	Information is processed & meaningful data.

It has no meaning on its own.	It helps in decision making.
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Q3. What is Database & DBMS?

- **Database**

Database is an organized collection of related data stored electronically for easy access & management.

A database helps reduce data duplication and makes information easy to retrieve. It is widely used in schools, banks, hospitals, businesses and many other organizations.

- **DBMS (Database Management System)**

DBMS is software that allows users to create, store, update, Retrieve and manage database. It also provides security, backup, and recovery features.

Example of DBMS include MySQL, Oracle, & Microsoft SQL Server.

Q4. What is RDBMS?

RDBMS is stand for “*Relational Database management System*”. RDBMS is type of *DBMS* that stores data in tables Consisting of rows & columns. These tables are connected using “Primary key & foreign key”. RDBMS reduce data duplication & maintains data integrity. It supports SQL for managing & querying data efficiently. Example include MySQL Oracle, PostgreSQL & SQL Server.

Q5. States at least 5 names of Database Management System (DBMS) or Software.

Some popular Database Management System are:

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- SQLite